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ABOUT ME

Second-year Electrical Engineering student at Universitas Gadjah Mada with a strong interest in IoT and AI development. Awarded the Paragon Scholarship 2025 for academic excellence and potential in leadership and community impact. Experienced in event organization, with proven leadership in managing 150 participants and coordinating large-scale initiatives. Passionate about leveraging technology to support sustainable innovation, with a long-term goal of contributing to net-zero emission efforts in Indonesia.

EDUCATION

Universitas Gadjah Mada	August 2025 - Present
Bachelor of Electrical Engineering	

SKILLS

Technical Skills:

- AI: Familiar with Machine Learning, Deep Learning, and Large Language Models (LLMs) at a using PyTorch.
- PCB Design: Familiar with PCB (Printed Circuit Board) design workflows using KiCad at an entry-level capacity.
- Web Development: Full-stack web developer experienced in building responsive, modern, and scalable web applications using Next.js, Tailwind CSS, and Payload CMS. Successfully completed several individual projects and collaborated in developing websites for organizational and event committees.
- Design: Canva, Figma & PicsArt (poster, infographics, and instagram puzzle feeds).

Soft Skills:

- Public Speaking: 3rd Winner at Da'i Competition Islamic Youth Festival STT Nurul Fikri 2023
- Leadership & Event Management: Led a 20-member team with 150+ participants.
- Languages: English (TOEFL Prediction Score 503-LIA Certificate) & Indonesian (Native)

ORGANIZATIONAL & WORK EXPERIENCE

Mathematics Teacher at Radian Edu Solution Yogyakarta	August 2025 – Present
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- Delivered mathematics instruction to students from elementary to senior high school levels, including intensive preparation for the UTBK, with a focus on Pengetahuan Kuantitatif and Penalaran Matematika subtests.
- Designed 5+ structured lesson plans (RPP) and syllabi tailored to the curriculum.
- Created tailored exercises and practice materials to reinforce understanding.

Staff of IT Pionir Gadjah Mada 2026	March 2026 – August 2026
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- Translating design mockups into high-quality, responsive code using Framework Next.js + Tailwind CSS.
- Optimizing web applications for maximum speed, scalability, and cross-browser compatibility.
- Collaborating with back-end teams to integrate front-end components with server-side logic and APIs.

- Developing interactive 3D web applications and games using modern web technologies (e.g., WebGL and Three.js).

Staff of Front-End Developer Website Development NESCO 2026

December 2025 – June 2026

- Translating design mockups into high-quality, responsive code using Framework Next.js + Tailwind CSS.
- Optimizing web applications for maximum speed, scalability, and cross-browser compatibility.
- Collaborating with back-end teams to integrate front-end components with server-side logic and APIs.

Staff of Front-End Developer Website Development Technocorner 2026

December 2025 – July 2026

- Translating design mockups into high-quality, responsive code using Framework Next.js + Tailwind CSS.
- Optimizing web applications for maximum speed, scalability, and cross-browser compatibility.
- Collaborating with back-end teams to integrate front-end components with server-side logic and APIs.

AWARDS & ACHIEVEMENTS

3rd Winner at Da'i Competition Islamic Youth Festival

December 2023

STT Nurul Fikri Jakarta

Grantee of Paragon Scholarship

November 2025

Paragon Technology and Innovation

COURSES

LIA Preparation Course for TOEFL Test

August 2023 - August 2024

I completed a one-year TOEFL preparation course at LIA, attending classes once a week. The program enhanced my proficiency in listening, structure, reading, and writing, contributing to a significant improvement in my TOEFL skills.

PROJECTS

Smartwatch Guardian With WellnessJourney: Teknologi Inovatif Sebagai Solusi Efektif Untuk Melindungi Generasi Z Dari Hipertensi

June 2024

- Developed the concept of a smartwatch prototype designed to monitor blood pressure, sleep quality, and physical activity.
- Built a digital prototype with 3+ health-monitoring features.
- Designed a digital application prototype featuring an interactive Wellness Quest ecosystem to drive long-term user engagement.

Electronic Team Intern at Gadjah Mada Robotic Team

November 2025

- Designed PCB layouts using KiCad and performed manual fabrication through heat transfer (ironing), drilling, and assembly to create functional boards.
- Executed precise wiring and electrical connections for robotic systems to ensure reliable hardware performance.

Personal Portfolio Website (<https://aqilakresnaarrafi.vercel.app/>)

November 2025

- Built a responsive personal portfolio using Next.js and Tailwind CSS to showcase projects and skills.
- Developed a simple interactive AI system based on Large Language Models (LLMs) using PyTorch to handle and respond to user queries.
- Deployed the website for public access, improving online visibility and personal branding.

Alzheimer's Disease Classification Using Multiple Instance Learning (MIL) and Deep Learning

July 2026

- Selected as a UGM contingent representative in Data Mining for GEMASTIK 2026 after securing 3rd place in UGM's internal selection.
- Developed a deep learning model for Alzheimer's disease classification from brain MRI images using the Multiple Instance Learning (MIL) approach.
- Implemented a ResNet18-based feature extractor and trained the model to classify multiple stages of dementia from MRI scan bags.
- Utilized a Kaggle brain MRI dataset with image augmentation performed using Roboflow to support model training and improve data diversity.

Real-Time AI Posture Monitor

September 2026

- Developed a real-time AI posture monitoring application using YOLO11 Pose and webcam-based computer vision to track shoulder tilt, head displacement, and torso inclination.
- Designed a posture scoring system from 0–100 with GOOD, FAIR, and BAD classifications, providing real-time visual feedback and corrective posture recommendations.
- Implemented personalized posture calibration, Exponential Moving Average smoothing, and continuous bad-posture detection to improve monitoring accuracy and stability.
- Generated session analytics in JSON format, including average posture score, posture category distribution, bad-posture duration, and frequently detected posture issues.